Vapor Pressure and Temperature

The relationship between vapor pressure and temperature can be understood in terms of molecules in the liquid escaping the attractive forces of the liquid to enter the gas phase.

Boiling Point and Vapor Pressure

When a liquid is heated in an open vessel, the ***vapor pressure*** will increase until it reaches the atmospheric pressure surrounding the liquid. When this happens, it is accompanied by the physical phenomena of boiling, as vapor bubbles form in the liquid and rise to the surface.

The ***boiling point*** of a liquid is "the temperature at which its vapor pressure is equal to the surrounding pressure."

Specific Gravity

Specific gravity of a liquid is "a ratio of the weight of a volume of the liquid to the weight of an equal volume of water at a designated temperature."

Specific gravity of a gas is "a ratio of the weight of a volume of the gas to the weight of an equal volume of air at a designated temperature and pressure."

Table 1: Specific Gravity

Parameter	Methane C1 (CH_4)	Ethane C2 (C_2H_6)	Propane C3 (C_3H_8)	Iso-Butane iC4 (C_4H_{10})	N-Butane nC4 (C_4H_{10})
Molecular Weight	16	30	44	58	58
Liquid Specific Gravity	0.3	0.36	0.51	0.56	0.58
Vapor Specific Gravity	0.55	1.04	1.52	2.01	2.01

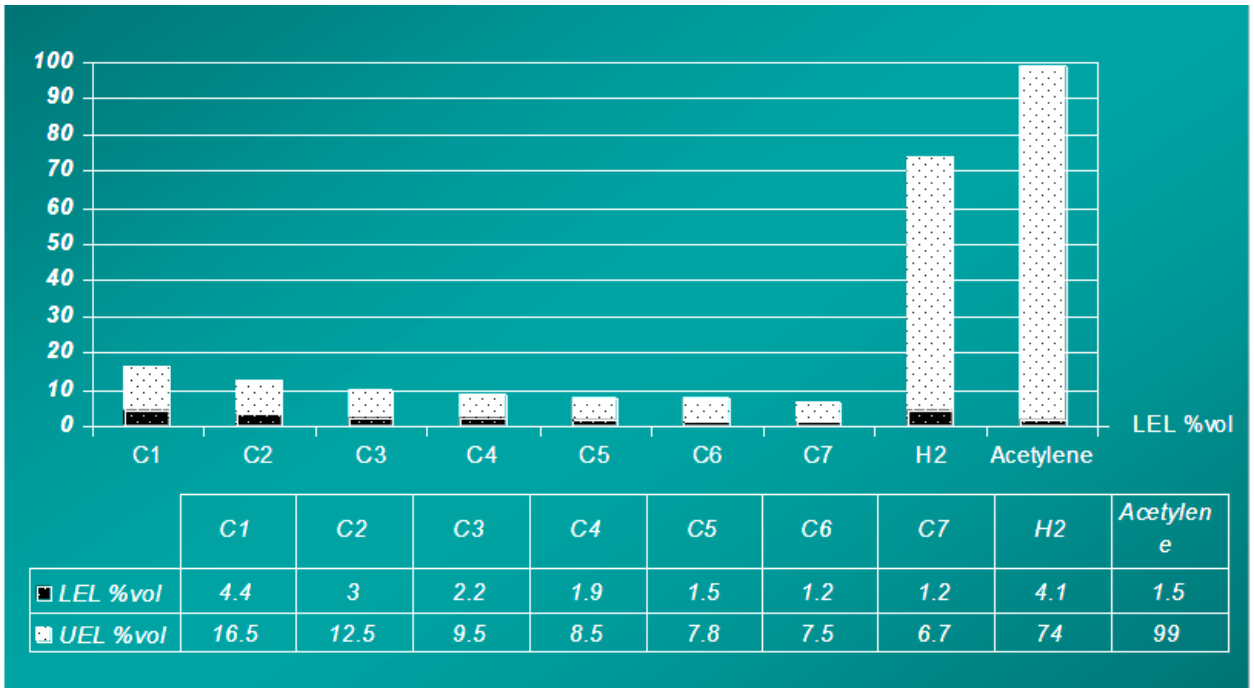

Figure 3: Flammability Limits of Hydrocarbons

Table 2: Physical Properties of Hydrocarbons

Parameter	C1 Methane	C2 Ethane	C3 Propane
Chemical Formula	**CH_4**	C_2H_6	$C3H_8$
Molecular Weight	16	30	44
Boiling Point	**–259.6°F**	–128.2°F	–43.6°F

All the above hydrocarbons have a general formula of CnH2n+2 and belong to the classification of hydrocarbons called “paraffins.”